

ERBIL INTL, Iraq

WMO: 406356

Lat: 36.233N

Lon: 43.967E

Elev: 1341

StdP: 14.00

Time Zone: 3.00 (E03)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.6	33.5	12.1	10.7	64.5	16.1	13.1	67.0	25.8	51.1	22.7	50.0	3.5	300	0.466

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.6	113.0	67.5	111.0	67.4	108.0	66.5	71.9	102.5	69.8	102.3	68.3	103.6	9.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.3	88.2	80.3	60.0	81.2	74.9	57.9	75.3	73.5	36.4	102.5	34.5	102.2	33.2	103.6	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.8	16.8	14.2	DB	26.2	115.7	3.0	2.5	24.0	117.5	22.2	119.0	20.5	120.4	18.4	122.3
			WB	25.1	75.0	2.8	4.6	23.1	78.3	21.5	81.0	20.0	83.5	18.0	86.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.8	46.8	50.0	55.8	64.9	76.4	88.3	94.1	93.7	84.8	72.9	58.3	50.1	
	DBStd	17.77	4.95	5.78	6.19	5.83	5.96	4.36	3.71	3.43	5.02	6.04	6.41	4.93	
	HDD50	272	121	65	20	0	0	0	0	0	0	10	56		
	HDD65	2032	565	421	291	70	2	0	0	0	9	213	461		
	CDD50	7497	21	64	198	447	820	1148	1368	1355	1045	711	259	61	
	CDD65	3781	0	0	5	67	357	698	903	890	595	254	12	0	
	CDH74	60736	0	2	73	919	5048	11549	16013	15418	8707	2839	160	8	
	CDH80	40403	0	0	5	260	2713	7845	11701	11108	5495	1262	13	1	
Wind	WSAvg	5.9	5.7	5.5	6.3	6.2	6.5	6.9	6.5	6.0	5.4	5.6	4.6	5.2	
Precipitation	PrecAvg	18.7	2.3	2.4	2.6	2.7	1.6	0.5	0.5	0.3	0.4	1.3	2.0	2.3	
	PrecMax	25.0	3.7	4.5	5.2	4.8	4.7	1.5	1.0	0.7	1.3	3.2	4.7	4.7	
	PrecMin	11.2	0.9	0.6	1.1	0.9	0.4	0.1	0.1	0.1	0.0	0.2	0.2	0.5	
	PrecStd	3.3	0.8	0.9	1.1	1.1	0.9	0.4	0.2	0.2	0.3	0.8	1.2	1.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.7	71.9	80.6	91.1	102.0	111.6	115.3	115.1	109.0	97.2	82.1	71.8	
		MCWB	49.7	53.3	58.5	59.6	63.9	67.0	67.8	68.2	66.4	63.3	58.2	52.8	
	2%	DB	61.2	68.2	75.5	86.2	98.7	109.0	113.2	113.1	105.9	93.3	78.7	66.2	
		MCWB	49.5	52.8	56.2	60.2	63.8	66.4	67.3	67.8	65.0	61.5	55.9	51.6	
	5%	DB	59.0	64.7	72.0	82.5	95.3	105.9	111.2	111.0	102.6	90.3	75.2	62.8	
		MCWB	49.4	52.1	55.0	59.9	62.9	65.7	67.2	67.8	64.9	60.7	54.8	50.0	
	10%	DB	56.8	61.0	69.4	79.2	93.1	103.6	109.0	108.6	100.1	87.7	71.5	60.4	
		MCWB	48.8	50.4	54.5	59.3	62.4	64.7	66.6	67.4	63.9	60.0	54.6	49.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	57.3	61.1	66.6	69.2	72.1	74.0	76.0	71.9	68.2	63.0	56.6
			MCDB	58.2	66.3	70.7	79.6	91.1	99.2	104.3	105.7	97.2	88.4	71.7	63.1
2%		WB	53.5	54.9	59.1	63.8	67.0	69.1	70.8	72.7	68.4	65.5	60.5	54.6	
		MCDB	57.5	63.4	70.6	78.2	89.9	99.3	104.6	103.5	96.7	82.4	69.0	60.9	
5%		WB	51.2	53.5	57.4	62.3	65.3	67.4	69.0	70.0	66.5	63.9	58.6	53.1	
		MCDB	56.8	61.6	69.1	75.7	87.8	101.3	106.5	104.8	97.8	80.9	68.4	59.2	
10%		WB	49.6	51.8	55.8	60.9	63.9	65.9	67.7	68.2	65.2	62.5	57.0	51.8	
		MCDB	54.9	59.5	66.9	73.6	85.9	100.1	106.5	106.3	96.9	78.6	67.4	57.5	
Mean Daily Temperature Range		5% DB	MDBR	17.4	18.9	20.6	24.8	27.2	28.3	28.6	28.8	27.9	24.0	21.9	18.3
			MCDBR	22.4	24.4	26.2	29.9	30.8	31.5	31.8	31.9	31.0	28.0	27.2	24.1
	MCWBR		13.1	12.5	12.4	12.2	12.1	11.7	12.1	12.5	12.7	13.3	14.3	13.9	
	5% WB	MCDBR	16.6	19.5	21.7	25.3	28.0	28.1	28.9	28.2	27.4	21.6	20.4	17.2	
		MCWBR	11.4	11.1	11.6	12.2	12.1	12.9	13.4	14.1	13.7	12.5	12.2	11.1	
Clear-Sky Solar Irradiance	taub	0.403	0.434	0.471	0.553	0.544	0.481	0.504	0.470	0.435	0.465	0.399	0.395		
	taud	2.057	1.987	1.918	1.757	1.797	1.940	1.892	1.976	2.054	1.964	2.115	2.090		
	Ebn at Noon	239	247	250	237	241	257	250	256	258	233	239	232		
	Edh at Noon	40	49	57	70	69	60	62	56	49	49	38	37		
All-Sky Solar Radiation	RadAvg	761	1031	1414	1753	2161	2551	2435	2214	1847	1292	939	704		
	RadStd	74	110	133	127	125	115	66	72	86	95	97	91		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (5 neighbors)	N/A	-3.22	N/A	+2.25	N/A	-0.68	N/A	N/A	N/A	N/A	

Nomenclature: See separate page